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Data Memory Initialized

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Region0 : 0x00 - 0x3F (R/W/X)

Region1 : 0x40 - 0x7F (R/X)

Region2 : 0x80 - 0xBF (No Access)

Default : 0xC0 - 0xFF (No Access)

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Instruction Memory Initialized

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PMP Test Program Loaded

[0] addi x5, x0, 0x80

[1] addi x6, x0, 0x40

[2] addi x7, x0, 0x00

[3] addi x8, x0, 0x0C0

[4] addi x1, x0, 1

[5] lw x1, 0(x7) - ALLOWED

[6] lw x2, 0(x5) - VIOLATION #1

[7] sw x1, 0(x6) - VIOLATION #2

[8] add x3, x2, x3

[9] lw x1, 0(x8) - VIOLATION #4

[10] jalr x1, x5, 0 - VIOLATION #3

[11] jal x0, 0x58 - HALT

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[0] PMP DENIED Addr=0xxx R=0 W=0 X=1 Perm=000

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[PMP] TEST PROGRAM LOADED

Expected: 4 PMP Violations

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[PASS] RESET DEASSERTED AT 110000 ps

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| TIME | PC | INSTRUCTION | ALU RESULT | STAT |

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| 130000 | 0x04 | ADDI x 6, x 0, 0x40 | 0x00000040 | P |

| | | WRITE x 6 = 0x00000000 | | |

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| REGS | | x0=0x00000000 | x1=0x00000000 | x2=0x00000000 | x3=0x00000000 |

| | | x4=0x00000000 | x5=0x00000000 | x6=0x00000040 | x7=0x00000000 |

| | | x8=0x00000000 | x9=0x00000000 | x10=0x00000000 | x11=0x00000000 |

| | | x12=0x00000000 | x13=0x00000000 | x14=0x00000000 | x15=0x00000000 |

| | | x16=0x00000000 | x17=0x00000000 | x18=0x00000000 | x19=0x00000000 |

| | | x20=0x00000000 | x21=0x00000000 | x22=0x00000000 | x23=0x00000000 |

| | | x24=0x00000000 | x25=0x00000000 | x26=0x00000000 | x27=0x00000000 |

| | | x28=0x00000000 | x29=0x00000000 | x30=0x00000000 | x31=0x00000000 |

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- - [WARN] REG MISMATCH: x 6 exp=0x00000040 act=0x00000000 - -

| 150000 | 0x08 | ADDI x 7, x 0, 0x00 | 0x00000000 | P |

| | | WRITE x 7 = 0x00000000 | | |

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| REGS |   | x0=0x00000000 | x1=0x00000000 | x2=0x00000000 | x3=0x00000000 |
|   |   | x4=0x00000000 | x5=0x00000000 | x6=0x00000040 | x7=0x00000000 |
|   |   | x8=0x00000000 | x9=0x00000000 | x10=0x00000000 | x11=0x00000000 |
|   |   | x12=0x00000000 | x13=0x00000000 | x14=0x00000000 | x15=0x00000000 |
|   |   | x16=0x00000000 | x17=0x00000000 | x18=0x00000000 | x19=0x00000000 |
|   |   | x20=0x00000000 | x21=0x00000000 | x22=0x00000000 | x23=0x00000000 |
|   |   | x24=0x00000000 | x25=0x00000000 | x26=0x00000000 | x27=0x00000000 |
|   |   | x28=0x00000000 | x29=0x00000000 | x30=0x00000000 | x31=0x00000000 |
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| 170000 | 0x0c | ADDI x 8, x 0, 0xc0 | 0x000000c0 | P |
|   |   | WRITE x 8 = 0x00000000 |   |   |
+-----+-----+-----+-----+
| REGS |   | x0=0x00000000 | x1=0x00000000 | x2=0x00000000 | x3=0x00000000 |
|   |   | x4=0x00000000 | x5=0x00000000 | x6=0x00000040 | x7=0x00000000 |
|   |   | x8=0x000000c0 | x9=0x00000000 | x10=0x00000000 | x11=0x00000000 |
|   |   | x12=0x00000000 | x13=0x00000000 | x14=0x00000000 | x15=0x00000000 |
|   |   | x16=0x00000000 | x17=0x00000000 | x18=0x00000000 | x19=0x00000000 |
|   |   | x20=0x00000000 | x21=0x00000000 | x22=0x00000000 | x23=0x00000000 |
|   |   | x24=0x00000000 | x25=0x00000000 | x26=0x00000000 | x27=0x00000000 |
|   |   | x28=0x00000000 | x29=0x00000000 | x30=0x00000000 | x31=0x00000000 |
+-----+-----+-----+-----+
- - [WARN] REG MISMATCH: x 8 exp=0x000000c0 act=0x00000000 - -
| 190000 | 0x10 | ADDI x 1, x 0, 0x01 | 0x00000001 | P |
|   |   | WRITE x 1 = 0x00000000 |   |   |
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| REGS |   | x0=0x00000000 | x1=0x00000001 | x2=0x00000000 | x3=0x00000000 |
|   |   | x4=0x00000000 | x5=0x00000000 | x6=0x00000040 | x7=0x00000000 |
|   |   | x8=0x000000c0 | x9=0x00000000 | x10=0x00000000 | x11=0x00000000 |
|   |   | x12=0x00000000 | x13=0x00000000 | x14=0x00000000 | x15=0x00000000 |
|   |   | x16=0x00000000 | x17=0x00000000 | x18=0x00000000 | x19=0x00000000 |
|   |   | x20=0x00000000 | x21=0x00000000 | x22=0x00000000 | x23=0x00000000 |
|   |   | x24=0x00000000 | x25=0x00000000 | x26=0x00000000 | x27=0x00000000 |
|   |   | x28=0x00000000 | x29=0x00000000 | x30=0x00000000 | x31=0x00000000 |
+-----+-----+-----+-----+
- - [WARN] REG MISMATCH: x 1 exp=0x00000001 act=0x00000000 - -
DATA MEM READ : Addr=0x001 Data=0x00000002 Time=190000
DATA MEM READ : Addr=0x000 Data=0x00000001 Time=190000
| 210000 | 0x14 | LW x 1, 0(x 7) | 0x00000000 | P |
|   |   | WRITE x 1 = 0x00000001 |   |   |
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| REGS |   | x0=0x00000000 | x1=0xxxxxxxxx | x2=0x00000000 | x3=0x00000000 |
|   |   | x4=0x00000000 | x5=0x00000000 | x6=0x00000040 | x7=0x00000000 |
|   |   | x8=0x000000c0 | x9=0x00000000 | x10=0x00000000 | x11=0x00000000 |
|   |   | x12=0x00000000 | x13=0x00000000 | x14=0x00000000 | x15=0x00000000 |
|   |   | x16=0x00000000 | x17=0x00000000 | x18=0x00000000 | x19=0x00000000 |
|   |   | x20=0x00000000 | x21=0x00000000 | x22=0x00000000 | x23=0x00000000 |
|   |   | x24=0x00000000 | x25=0x00000000 | x26=0x00000000 | x27=0x00000000 |
|   |   | x28=0x00000000 | x29=0x00000000 | x30=0x00000000 | x31=0x00000000 |
+-----+-----+-----+-----+
- - [WARN] REG MISMATCH: x 1 exp=0xxxxxxxxx act=0x00000001 - -
DATA MEM READ : Addr=0x080 Data=0xdeadbeef Time=210000
[210000] PMP DENIED Addr=0x80 R=1 W=0 X=0 Perm=000

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| 230000 | 0x18 | LW x 2, 0(x 5) | 0x00000080 | P |
| | | WRITE x 2 = 0x00000000 | | |

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REGS		x0=0x00000000	x1=0xxxxxxxx	x2=0xxxxxxxx	x3=0x00000000
		x4=0x00000000	x5=0x00000000	x6=0x00000040	x7=0x00000000
		x8=0x000000c0	x9=0x00000000	x10=0x00000000	x11=0x00000000
		x12=0x00000000	x13=0x00000000	x14=0x00000000	x15=0x00000000
		x16=0x00000000	x17=0x00000000	x18=0x00000000	x19=0x00000000
		x20=0x00000000	x21=0x00000000	x22=0x00000000	x23=0x00000000
		x24=0x00000000	x25=0x00000000	x26=0x00000000	x27=0x00000000
		x28=0x00000000	x29=0x00000000	x30=0x00000000	x31=0x00000000
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- - [WARN] REG MISMATCH: x 2 exp=0xxxxxxxx act=0x00000000 - -

- - [DENY] #0: Load from Region2 (0x80) - - -

[230000] PMP DENIED Addr=0x80 R=0 W=1 X=0 Perm=000

[230000] PMP DENIED Addr=0x40 R=0 W=1 X=0 Perm=101

| 250000 | 0x1c | SW x 1, 0(x 6) | 0x00000040 | P |

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REGS		x0=0x00000000	x1=0xxxxxxxx	x2=0xxxxxxxx	x3=0x00000000
		x4=0x00000000	x5=0x00000000	x6=0x00000040	x7=0x00000000
		x8=0x000000c0	x9=0x00000000	x10=0x00000000	x11=0x00000000
		x12=0x00000000	x13=0x00000000	x14=0x00000000	x15=0x00000000
		x16=0x00000000	x17=0x00000000	x18=0x00000000	x19=0x00000000
		x20=0x00000000	x21=0x00000000	x22=0x00000000	x23=0x00000000
		x24=0x00000000	x25=0x00000000	x26=0x00000000	x27=0x00000000
		x28=0x00000000	x29=0x00000000	x30=0x00000000	x31=0x00000000
+-----+-----+-----+-----+

- - [DENY] #1: Store to Region1 (0x40) - - -

| 270000 | 0x20 | ADD x 3, x 2, x 3 | 0x00000040 | P |

| | | WRITE x 3 = 0x00000000 | | |

+-----+-----+-----+-----+
REGS		x0=0x00000000	x1=0xxxxxxxx	x2=0xxxxxxxx	x3=0xxxxxxxx
		x4=0x00000000	x5=0x00000000	x6=0x00000040	x7=0x00000000
		x8=0x000000c0	x9=0x00000000	x10=0x00000000	x11=0x00000000
		x12=0x00000000	x13=0x00000000	x14=0x00000000	x15=0x00000000
		x16=0x00000000	x17=0x00000000	x18=0x00000000	x19=0x00000000
		x20=0x00000000	x21=0x00000000	x22=0x00000000	x23=0x00000000
		x24=0x00000000	x25=0x00000000	x26=0x00000000	x27=0x00000000
		x28=0x00000000	x29=0x00000000	x30=0x00000000	x31=0x00000000
+-----+-----+-----+-----+

- - [WARN] REG MISMATCH: x 3 exp=0xxxxxxxx act=0x00000000 - -

DATA MEM READ : Addr=0x100 Data=0x00000000 Time=270000

DATA MEM READ : Addr=0x0c0 Data=0xffffffff Time=270000

[270000] PMP DENIED Addr=0xc0 R=1 W=0 X=0 Perm=000

| 290000 | 0x24 | LW x 1, 0(x 8) | 0x000000c0 | P |

| | | WRITE x 1 = 0x00000001 | | |

+-----+-----+-----+-----+
REGS		x0=0x00000000	x1=0xxxxxxxx	x2=0xxxxxxxx	x3=0xxxxxxxx
		x4=0x00000000	x5=0x00000000	x6=0x00000040	x7=0x00000000
		x8=0x000000c0	x9=0x00000000	x10=0x00000000	x11=0x00000000
		x12=0x00000000	x13=0x00000000	x14=0x00000000	x15=0x00000000
		x16=0x00000000	x17=0x00000000	x18=0x00000000	x19=0x00000000
		x20=0x00000000	x21=0x00000000	x22=0x00000000	x23=0x00000000
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|   |   |x24=0x00000000|x25=0x00000000|x26=0x00000000|x27=0x00000000|
|   |   |x28=0x00000000|x29=0x00000000|x30=0x00000000|x31=0x00000000|
+-----+-----+-----+-----+-----+-----+
- - [WARN] REG MISMATCH: x 1 exp=0xxxxxxxx act=0x00000001 - -
- - [DENY] #2: Load from Default (0xc0) - - -
| 310000 | 0x28 | JALR x 1, x 5, 0 | 0x00000080 | P |
|   |   | WRITE x 1 = 0x00000000 |   |   |
+-----+-----+-----+-----+-----+-----+
| REGS |   |x0=0x00000000|x1=0x0000002c|x2=0xxxxxxxx|x3=0xxxxxxxx|
|   |   |x4=0x00000000|x5=0x00000000|x6=0x00000040|x7=0x00000000|
|   |   |x8=0x000000c0|x9=0x00000000|x10=0x00000000|x11=0x00000000|
|   |   |x12=0x00000000|x13=0x00000000|x14=0x00000000|x15=0x00000000|
|   |   |x16=0x00000000|x17=0x00000000|x18=0x00000000|x19=0x00000000|
|   |   |x20=0x00000000|x21=0x00000000|x22=0x00000000|x23=0x00000000|
|   |   |x24=0x00000000|x25=0x00000000|x26=0x00000000|x27=0x00000000|
|   |   |x28=0x00000000|x29=0x00000000|x30=0x00000000|x31=0x00000000|
+-----+-----+-----+-----+-----+-----+
- - [WARN] REG MISMATCH: x 1 exp=0x0000002c act=0x00000000 - -
[310000] PMP DENIED Addr=0x80 R=0 W=0 X=1 Perm=000
- - [DENY] #3: Execute from Region2 (PC=0x80) - - -
[330000] PMP DENIED Addr=0x84 R=0 W=0 X=1 Perm=000
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[PASS] PMP TEST PASSED
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All 4 PMP violation types detected:
[PASS] Load from Region2 : DETECTED
[PASS] Store to Region1 : DETECTED
[PASS] Execute from Region2 : DETECTED
[PASS] Load from Default : DETECTED
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Total Violations Count = 4
Cycles Executed = 12
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[350000] PMP DENIED Addr=0x88 R=0 W=0 X=1 Perm=000
$finish called from file "design.sv", line 958.
$finish at simulation time 370000
V C S Simulation Report
Time: 370000 ps
CPU Time: 0.660 seconds; Data structure size: 0.0Mb
Thu Oct 1 15:52:00 2026
Finding VCD file...
./waveform.vcd
[2026-10-01 19:52:00 UTC] Opening EPWave...
Done

```